

Introduction to Differential Equations

DAY-1

Topics to cover:

- ➔ Introduction to Differential Equations (DE)
- ➔ Type of Solutions
- ➔ Mathematical Modeling

Introduction

Definition: An equation containing the derivatives of one or more unknown functions (or dependent variables), with respect to one or more independent variables, is said to be a differential equation (DE).

A differential equation of y :

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0.$$

Here the unknown of the equation is some differentiable (hence continuous) function $y = \varphi(x)$ and we solve the equation to find the function $\varphi(x)$.

[In the equation $x^2 + 5x + 4 = 0$ the unknown is x and we solve the equation to find the values of x .]

Modeling:

A. Population Growth Dynamics

In a country,

the rate of change of the number of Population is directly proportional to the number of Population.

Let P = number of Population in a country. Then

$$\frac{dP}{dt} \propto P.$$

If the constant k is the constant of proportionality, then $k > 0$ (as the rate of change is positive),

$$\frac{dP}{dt} = kP$$

B. Radioactive Decay

The nucleus of an atom consists of combinations of protons and neutrons. Many of these combinations of protons and neutrons are unstable—that is, the atoms decay or transmute into atoms of another substance. Such nuclei are said to be radioactive. For example, over time the highly radioactive radium, Ra-226, transmutes into the radioactive gas radon, Rn-222. To model the phenomenon of radioactive decay, it is assumed that the rate $\frac{dA}{dt}$ at which the nuclei of a substance decay is proportional to the amount (more precisely, the number of nuclei) $A(t)$ of the substance remaining at time t :

$$\frac{dA}{dt} \propto A.$$

If the constant k is the constant of proportionality, then $k < 0$ (as the rate of change is negative),

$$\frac{dA}{dt} = kA$$

A few examples of DE:

- | | | | | |
|-----|---|-------------|---------------|--------|
| (1) | $\frac{dy}{dx} + 5y = e^x$ | ; Ordinary, | First Order, | Linear |
| (2) | $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 6y = 0$ | ; Ordinary, | Second Order, | Linear |
| (3) | $\frac{dx}{dt} + \frac{dy}{dt} = 2x + y$ | ; Ordinary, | First Order, | Linear |
| (4) | $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ | ; Partial, | Second Order, | Linear |
| (5) | $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - 2 \frac{\partial u}{\partial t}$ | ; Partial, | Second Order, | Linear |
| (6) | $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ | ; Partial, | First Order, | Linear |

Classification of Differential Equations:

- A. Classification by Type
- B. Classification by Order
- C. Classification by Linearity

A. Classification by Type

Ordinary: If a differential equation contains only ordinary derivatives of one or more unknown functions with respect to a single independent variable, it is said to be an ordinary differential equation (ODE).

Partial: An equation involving partial derivatives of one or more unknown functions of two or more independent variables is called a partial differential equation (PDE). Our first example illustrates several of each type of differential equation.

B. Classification by Order

The order of a differential equation (either ODE or PDE) is the order of the highest derivative in the equation.

For Example, a second-order ordinary differential equation:

$$\overset{\text{second order} \downarrow}{\frac{d^2 y}{dx^2}} + 5 \overset{\downarrow \text{first order}}{\left(\frac{dy}{dx}\right)^3} - 4y = e^x$$

First-order ordinary differential equations are occasionally written in differential form

$$M(x, y)dx + N(x, y)dy = 0.$$

For example, if we assume that y denotes the dependent variable in

$$(y - x)dx + 4xdy = 0,$$

then $y' = \frac{dy}{dx}$, so by dividing by the differential dx , we get the alternative form

$$4xy' + y = x.$$

In symbols, we can express an n th-order ordinary differential equation in one dependent variable by the general form

$$F(x, y, y', \dots, y^{(n)}) = 0, \quad \dots \dots \dots (1)$$

where F is a real-valued function of $n + 2$ variables: $x, y, y', \dots, y^{(n)}$.

For both practical and theoretical reasons we shall also make the assumption hereafter that it is possible to solve an ordinary differential equation in the form (1) uniquely for the highest derivative $y^{(n)}$ in terms of the remaining $n + 1$ variables. The differential equation

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}),$$

where f is a real-valued continuous function, is referred to as the normal form of (1).

C. Classification by Linearity

The form of an n^{th} order **Linear** ordinary differential equation:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \dots \dots \dots (2)$$

Since y is the dependent variable, so are all higher order derivatives of y , that is, $y', y'', \dots, y^{(n)}$ all are dependent variables.

An n^{th} order ordinary differential equation of type (2) is said to be linear, where

- (i) The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of first degree, that is, the power of each term involving y is 1.
- (ii) The coefficients $a_0, a_1, \dots, a_n$ of $y, y', y'', \dots, y^{(n)}$ depend at most on the independent variable x .
- (iii) The right-hand side function is ALWAYS a function of the independent variable x .

Note: A nonlinear ordinary differential equation is simply one that is not linear.

Nonlinear functions of the dependent variable or its derivatives, such as $\sin y$ or e^y , cannot appear in a linear equation.

Example of non-linear equations:

1. $(1 - y)y' + 2y = e^x$
2. $\frac{d^2 y}{dx^2} + \sin y = 0$
3. $\frac{d^4 y}{dx^4} + y^2 = \cos x$
4. $\frac{d^2 y}{dx^2} + y \sin x = \ln y$
5. $\frac{d^2 y}{dx^2} + y \frac{dy}{dx} = x^2$

Solutions of Differential Equations:

DEFINITION 1.1.2 Solution of an ODE

Any function ϕ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a **solution** of the equation on the interval.

DEFINITION 1.1.3 Implicit Solution of an ODE

A relation $G(x, y) = 0$ is said to be an **implicit solution** of an ordinary differential equation (4) on an interval I , provided that there exists at least one function ϕ that satisfies the relation as well as the differential equation on I .

Initial-Value Problems:

Consider the n^{th} order **Linear** ordinary differential equation:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad \dots (2),$$

where the initial value of x is $x = x_0$ and the values of all dependent variables $y, y', y'', \dots, y^{(n-1)}$ at $x = x_0$ are:

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad y''(x_0) = y_2, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}.$$

Then, the initial-value problem (IVP) of the n^{th} order **Linear** ordinary differential equation is given by:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

subject to the initial conditions (IC):

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad y''(x_0) = y_2, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}.$$

Note that an initial-value problem (IVP) gives a particular solution. If $y = \varphi(x)$ is a solution of the initial-value problem

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

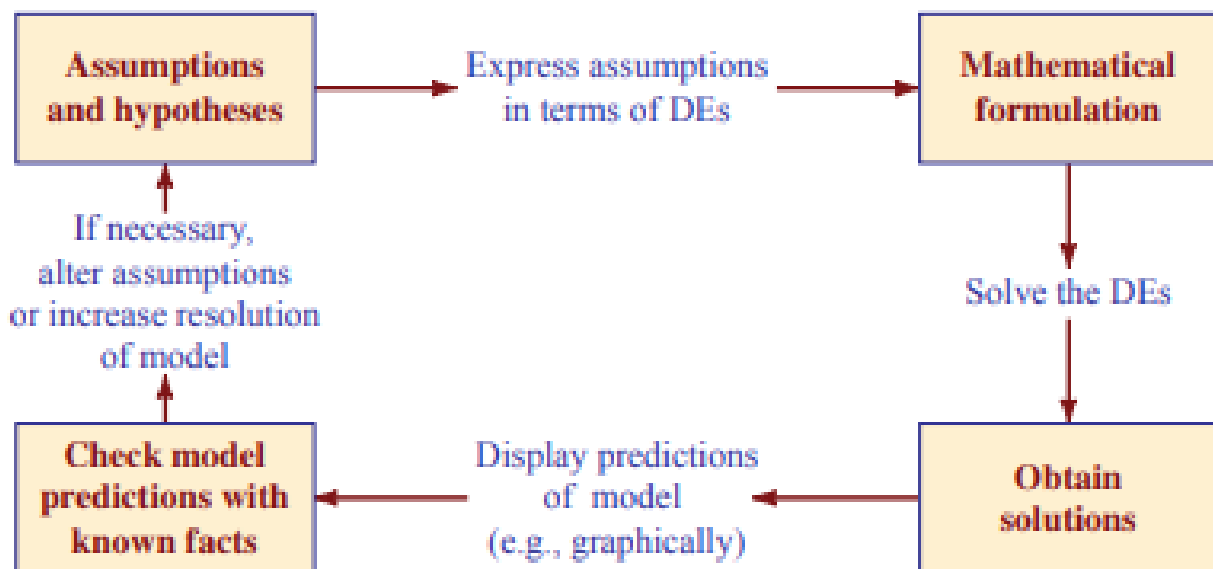
subject to the initial conditions (IC):

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad y''(x_0) = y_2, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1} ;$$

then the graph of the solution $y = \varphi(x)$ contains the point (x_0, y_0) .

DIFFERENTIAL EQUATIONS AS MATHEMATICAL MODELS

Steps in the modeling process with differential equations:



Study the following Models:

≡ Spread of a Disease A contagious disease—for example, a flu virus—is spread throughout a community by people coming into contact with other people. Let $x(t)$ denote the number of people who have contracted the disease and $y(t)$ denote the number of people who have not yet been exposed. It seems reasonable to assume that the rate dx/dt at which the disease spreads is proportional to the number of encounters, or *interactions*, between these two groups of people. If we assume that the number of interactions is jointly proportional to $x(t)$ and $y(t)$ —that is, proportional to the product xy —then

$$\frac{dx}{dt} = kxy, \quad (4)$$

where k is the usual constant of proportionality. Suppose a small community has a fixed population of n people. If one infected person is introduced into this community, then it could be argued that $x(t)$ and $y(t)$ are related by $x + y = n + 1$. Using this last equation to eliminate y in (4) gives us the model

$$\frac{dx}{dt} = kx(n + 1 - x). \quad (5)$$

An obvious initial condition accompanying equation (5) is $x(0) = 1$.

≡ Mixtures The mixing of two salt solutions of differing concentrations gives rise to a first-order differential equation for the amount of salt contained in the mixture. Let us suppose that a large mixing tank initially holds 300 gallons of brine (that is, water in which a certain number of pounds of salt has been dissolved). Another brine solution is pumped into the large tank at a rate of 3 gallons per minute; the concentration of the salt in this inflow is 2 pounds per gallon. When the solution in the tank is well stirred, it is pumped out at the same rate as the entering solution. See Figure 1.3.2. If $A(t)$ denotes the amount of salt (measured in pounds) in the tank at time t , then the rate at which $A(t)$ changes is a net rate:

$$\frac{dA}{dt} = \left(\begin{array}{c} \text{input rate} \\ \text{of salt} \end{array} \right) - \left(\begin{array}{c} \text{output rate} \\ \text{of salt} \end{array} \right) = R_{in} - R_{out}.$$

The input rate R_{in} at which salt enters the tank is the product of the inflow concentration of salt and the inflow rate of fluid. Note that R_{in} is measured in pounds per minute:

$$\begin{array}{ccc}
 \begin{array}{c} \text{concentration} \\ \text{of salt} \\ \text{in inflo} \\ \downarrow \end{array} & \begin{array}{c} \text{input rate} \\ \text{of brine} \\ \downarrow \end{array} & \begin{array}{c} \text{input rate} \\ \text{of salt} \\ \downarrow \end{array} \\
 R_{in} = (2 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = (6 \text{ lb/min}).
 \end{array}$$

Now, since the solution is being pumped out of the tank at the same rate that it is pumped in, the number of gallons of brine in the tank at time t is a constant 300 gallons. Hence the concentration of the salt in the tank as well as in the outflow is $c(t) = A(t)/300$ lb/gal, so the output rate R_{out} of salt is

$$\begin{array}{ccc}
 \begin{array}{c} \text{concentration} \\ \text{of salt} \\ \text{in outflo} \\ \downarrow \end{array} & \begin{array}{c} \text{output rate} \\ \text{of brine} \\ \downarrow \end{array} & \begin{array}{c} \text{output rate} \\ \text{of salt} \\ \downarrow \end{array} \\
 R_{out} = \left(\frac{A(t)}{300} \text{ lb/gal} \right) \cdot (3 \text{ gal/min}) = \frac{A(t)}{100} \text{ lb/min}.
 \end{array}$$

Then we get,

$$\begin{aligned}
 \frac{dA}{dt} &= R_{in} - R_{out} = 6 - \frac{1}{100} A \\
 \therefore \frac{dA}{dt} + \frac{1}{100} A &= 6
 \end{aligned}$$

Note: Here the general input and output rates of the brine solutions are equal.



*****Practice exercises → Section 1.3: Exercise 10-12.

Draining a Tank In hydrodynamics, **Torricelli's law** states that the speed v of efflux of water through a sharp-edged hole at the bottom of a tank filled to a depth h is the same as the speed that a body (in this case a drop of water) would acquire in falling freely from a height h —that is, $v = \sqrt{2gh}$, where g is the acceleration due to gravity. This last expression comes from equating the kinetic energy $\frac{1}{2}mv^2$ with the potential energy mgh and solving for v . Suppose a tank filled with water is allowed to drain through a hole under the influence of gravity. We would like to find the depth h of water remaining in the tank at time t . Consider the tank shown in Figure 1.3.3. If the area of the hole is A_h (in ft^2) and the speed of the water leaving the tank is $v = \sqrt{2gh}$ (in ft/s), then the volume of water leaving the tank per second is $A_h\sqrt{2gh}$ (in ft^3/s). Thus if $V(t)$ denotes the volume of water in the tank at time t , then

$$\frac{dV}{dt} = -A_h\sqrt{2gh}, \quad (9)$$

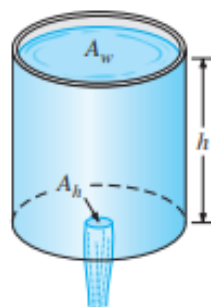


FIGURE 1.3.3 Draining tank

where the minus sign indicates that V is decreasing. Note here that we are ignoring the possibility of friction at the hole that might cause a reduction of the rate of flow there. Now if the tank is such that the volume of water in it at time t can be written $V(t) = A_w h$, where A_w (in ft^2) is the *constant* area of the upper surface of the water (see Figure 1.3.3), then $dV/dt = A_w dh/dt$. Substituting this last expression into (9) gives us the desired differential equation for the height of the water at time t :

$$\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2gh}. \quad (10)$$

It is interesting to note that (10) remains valid even when A_w is not constant. In this case we must express the upper surface area of the water as a function of h —that is, $A_w = A(h)$. See Problem 14 in Exercises 1.3.